SELECT '--- A. Generated columns derive grade from raw scores ---' AS test;
SELECT registration_id, ca_score, exam_score, total_score, grade, grade_point
FROM results ORDER BY registration_id;

	registration_id ▲ 1	ca_score	exam_score	total_score	grade	grade_point
Edit Copy Delete	1	32.00	45.50	77.50	A	4.00
Edit Copy Delete	2	25.00	45.00	70.00	AB	3.50
Edit Copy Delete	3	15.00	20.00	35.00	F	0.00

SELECT '--- B. UNIQUE: duplicate matric_no (EXPECT ERROR 1062) ---' AS test;
INSERT INTO students (programme_id, matric_no, first_name, last_name, gender, date_of_birth, email, admission_year, academic_level)
VALUES (1, 'FPE/ND/CST/26/001', 'Dup', 'Licate', 'Male', '1990-01-01', 'dup@stu.example', 2026, 100);

Error

SQL query: Copy

```
INSERT INTO students (programme_id, matric_no, first_name, last_name, gender,
                      date_of_birth, email, admission_year, academic_level)
VALUES (1, 'FPE/ND/CST/26/001', 'Dup', 'Licate', 'Male', '1990-01-01',
        'dup@stu.example', 2026, 100);
```

MySQL said:

```
#1062 - Duplicate entry 'FPE/ND/CST/26/001' for key 'uq_students_matric'
```

SELECT '--- C. UNIQUE: same student registers twice for one offering (EXPECT ERROR 1062) ---' AS test;
INSERT INTO course_registrations (student_id, offering_id) VALUES (1, 1);

Error

SQL query: Copy

```
INSERT INTO course_registrations (student_id, offering_id) VALUES (1, 1);
```

MySQL said:

```
#1062 - Duplicate entry '1-1' for key 'uq_registration_student_offering'
```

SELECT '--- D. FK: student with non-existent programme (EXPECT ERROR 1452) ---' AS test;
INSERT INTO students (programme_id, matric_no, first_name, last_name, gender,
date_of_birth, email, admission_year, academic_level)
VALUES (999, 'XXX/26/999', 'No', 'Programme', 'Male', '1990-01-01',

'noprog@stu.example', 2026, 100);

Error

SQL query: Copy

```
INSERT INTO students (programme_id, matric_no, first_name, last_name, gender,
                      date_of_birth, email, admission_year, academic_level)
VALUES (999, 'XXX/26/999', 'No', 'Programme', 'Male', '1990-01-01',
        'noprog@stu.example', 2026, 100);
```

MySQL said:

```
#1452 - Cannot add or update a child row: a foreign key constraint fails (`sarms_db`.`students`, CONSTRAINT `fk_students_programme` FOREIGN KEY (`programme_id`) REFERENCES `programmes` (`programme_id`) ON UPDATE CASCADE)
```

SELECT '--- E. CHECK: exam_score above 60 (EXPECT ERROR 4025 MariaDB / 3819 MySQL) ---' AS test;
INSERT INTO results (registration_id, ca_score, exam_score, entered_by)
VALUES (4, 20.00, 75.00, 3);

Error

SQL query: Copy

```
INSERT INTO results (registration_id, ca_score, exam_score, entered_by)
VALUES (4, 20.00, 75.00, 3);
```

MySQL said:

```
#4025 - CONSTRAINT `ck_results_exam` failed for `sarms_db`.`results`
```

SELECT '--- F. CHECK: credit_unit outside 1-6 (EXPECT ERROR) ---' AS test;
INSERT INTO courses (department_id, course_code, course_title, credit_unit, level, semester_type)
VALUES (1, 'CSC999', 'Bad Credit Course', 9, 100, 'First');

Error

SQL query: Copy

```
INSERT INTO courses (department_id, course_code, course_title, credit_unit, level, semester_type)
VALUES (1, 'CSC999', 'Bad Credit Course', 9, 100, 'First');
```

MySQL said:

```
#4025 - CONSTRAINT `ck_courses_credit` failed for `sarms_db`.`courses`
```

SELECT '--- H. CHECK: Student user without student_id (EXPECT ERROR) ---' AS test;
INSERT INTO users (username, password_hash, role)
VALUES ('orphan.student', 'hash', 'Student');

Error

SQL query: Copy

```
INSERT INTO users (username, password_hash, role)
VALUES ('orphan.student', 'hash', 'Student');
```

MySQL said:

```
#4025 - CONSTRAINT `ck_users_role_link` failed for `sarms_db`.`users`
```

SELECT '--- I. UNIQUE: a second result for the same registration (EXPECT ERROR 1062) ---' AS test;
INSERT INTO results (registration_id, ca_score, exam_score, entered_by)
VALUES (1, 10.00, 10.00, 3);

Error

SQL query: Copy

```
INSERT INTO results (registration_id, ca_score, exam_score, entered_by)
VALUES (1, 10.00, 10.00, 3);
```

MySQL said:

```
#1062 - Duplicate entry '1' for key 'uq_results_registration'
```

SELECT '--- J. RESTRICT: delete a department that still has dependent rows (EXPECT ERROR 1451) ---' AS test;
DELETE FROM departments WHERE department_id = 1;

Error

SQL query: Copy

```
DELETE FROM departments WHERE department_id = 1;
```

MySQL said:

```
#1451 - Cannot delete or update a parent row: a foreign key constraint fails (`sarms_db`.`courses`, CONSTRAINT `fk_courses_department`
FOREIGN KEY (`department_id`) REFERENCES `departments` (`department_id`) ON UPDATE CASCADE)
```

SELECT '--- K. NOT NULL: result without entered_by (EXPECT ERROR) ---' AS test;
INSERT INTO results (registration_id, ca_score, exam_score)
VALUES (4, 20.00, 20.00);

Error

SQL query: Copy

```
INSERT INTO results (registration_id, ca_score, exam_score)
VALUES (4, 20.00, 20.00);
```

MySQL said:

```
#1452 - Cannot add or update a child row: a foreign key constraint fails (`sarms_db`.`results`, CONSTRAINT `fk_results_entered_by` FOREIGN KEY (`entered_by`) REFERENCES `users` (`user_id`) ON UPDATE CASCADE)
```

SELECT '--- L. Valid insert succeeds and grade is derived (EXPECT SUCCESS) ---' AS test;
INSERT INTO results (registration_id, ca_score, exam_score, entered_by)
VALUES (4, 38.00, 52.00, 3);
SELECT registration_id, total_score, grade, grade_point
FROM results WHERE registration_id = 4;

1 row inserted.
Inserted row id: 6 (Query took 0.0053 seconds.)

```
INSERT INTO results (registration_id, ca_score, exam_score, entered_by) VALUES (4, 38.00, 52.00, 3);
```

[Edit inline] [Edit] [Create PHP code]

Showing rows 0 - 0 (1 total, Query took 0.0008 seconds.)

```
SELECT registration_id, total_score, grade, grade_point FROM results WHERE registration_id = 4;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 Filter rows: Search this table

Extra options

	registration_id	total_score	grade	grade_point
Edit Copy Delete	4	90.00	A	4.00

SELECT '--- M. CHECK: academic_level 700 is not an ND/HND level (EXPECT ERROR) ---' AS test;

```
INSERT INTO students (programme_id, matric_no, first_name, last_name, gender,
                      date_of_birth, email, admission_year, academic_level)
VALUES (1, 'FPE/ND/CST/26/099', 'Bad', 'Level', 'Male', '2006-01-01',
        'badlevel@stu.fpe.example', 2026, 700);
```

Error

SQL query: Copy

```
INSERT INTO students (programme_id, matric_no, first_name, last_name, gender,
                      date_of_birth, email, admission_year, academic_level)
VALUES (1, 'FPE/ND/CST/26/099', 'Bad', 'Level', 'Male', '2006-01-01',
        'badlevel@stu.fpe.example', 2026, 700);
```

MySQL said:

```
#4025 - CONSTRAINT `ck_students_level` failed for `sarms_db`.`students`
```

SELECT '--- N. Transcript-style join (all FKs resolve) ---' AS test;

```
SELECT s.matric_no, CONCAT(s.first_name,' ',s.last_name) AS student,
       c.course_code, c.credit_unit, r.total_score, r.grade, r.grade_point
FROM results r
JOIN course_registrations cr ON cr.registration_id = r.registration_id
JOIN students s              ON s.student_id       = cr.student_id
JOIN course_offerings o      ON o.offering_id      = cr.offering_id
JOIN courses c               ON c.course_id        = o.course_id
ORDER BY s.matric_no, c.course_code;
```

matric_no ▲ 1	student	course_code	credit_unit	total_score	grade	grade_point
FPE/ND/ACC/26/005	Ngozi Obi	ACC111	4	90.00	A	4.00
FPE/ND/CST/25/010	Tunde Ajayi	COM211	3	35.00	F	0.00
FPE/ND/CST/26/001	Chidi Nwosu	COM111	3	77.50	A	4.00
FPE/ND/CST/26/002	Amina Yusuf	COM111	3	70.00	AB	3.50